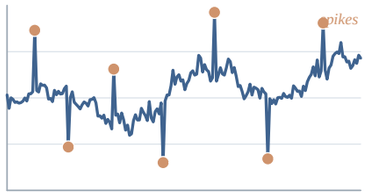
From Raw CSI to Recognition: Preprocessing and Model Effects

Different preprocessing choices preserve different signal representations and can change recognition outcomes

Stage I: Raw CSI Observation

Wireless sensing

Raw CSI waveform



Motion response + task-irrelevant interference

Interference sources

- Random noise and outliers
- Phase errors and offsets
- Hardware drift / baseline shift

The observed waveform is visibly jagged

with spikes, jitter, offsets, and slow drift

Stage II: Alternative Preprocessing

P1 Artifact-aware pipeline

Outlier removal → Phase calibration → Z-score scaling

P2 Trend-focused pipeline

Wavelet denoising → Linear detrending → Min-max scaling

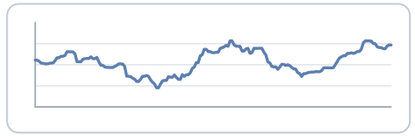
P3 Canonicalization pipeline

Moving-average smoothing → Phase sanitization → Resample & standardize

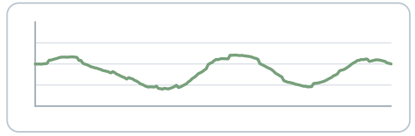
Each pipeline removes or preserves different signal characteristics

Stage III: CSI Representations

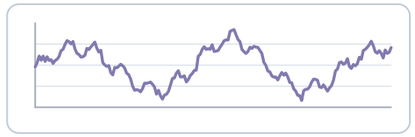
Representation R1 *detail-preserving*



Representation R2 *trend-emphasized*



Representation R3 *phase-aligned*



Stage IV: Recognition Outcomes

Controlled comparison

Same model M / data split / training settings

Model M → **Outcome 1** — **Class A** — *high confidence* — *illustrative*

Model M → **Outcome 2** — **Class A** — *medium confidence* — *illustrative*

Model M → **Outcome 3** — **Class B** — *medium confidence* — *illustrative*

Additional model-type effect

CNN | LSTM | Transformer

Recognition capability further modulates the outcomes

Input | Representation formation | Retained CSI characteristics | Recognition variability